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15 April 1971

MILITARY SPECIFICATION

POWER SUPPLY: 9342815 (KLYSTRON)

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. **This specification covers a power supply, hereinafter called klystron power supply, used in Test Set, Subassembly MX-8049/TPM-22, part of Radar Test Set AN/TPM-22A, and in Receiver-Transmitter, Radar RT-860/VPS-2, part of Radar Set AN/VPS-2 (see** 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this **specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and** Standards (DODISS) and **supplement thereto, cited In the** solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial Comments (recommendations, additions, deletions) and any pertinent data *which may be of use in improving* this document should be addressedto: **Commander, U.S Army** ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NewJersey07806-5000by using the self-addressed Standardization Document appearing at the end of this document or by **letter.**

AMSC **N/A** **FSC** 1285

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified. the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

Us. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9342815 - Klystron power supply

SPECIAL PACKAGING INSTRUCTION

PS 9342815 - Packaging of klystron power supply

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. **The components and assemblies** shall comply with all requirements specified on Drawing 9342815, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the klystron power supply shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The klystron power supply shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 *to* **15 hertz**	**0.06 inches double amplitude**
15 to 25 hertz	**0.04 inches double amplitude**
25 to 55 hertz	**0.02 inches double amplitude**

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 **Shock. The klystron power supply shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.**

3.3.3 Altitude.

3.3.3.1 Operating. The klystron power supply shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 **feet above sea level** *(20.6* **inches of Hg).**

3.3.3.2 **Non-operating. The klystron power supply shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).**

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3.3.4 **Temperature.**

3.3.4.1 Operating. **The klystron power supply shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.**

3.3.4.2 **Non-operating** (storage). **The klystron power supply shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned** to room ambient.

3.3.5 Humidity (operating) The klystron power supply shall meet the requirements of 3.4 and 3.5 during exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the klystron power supply.

3.5 Performance. (See High Voltage Warning 6.5)

3.5.1 Interlock switch. Interlock circuit between J1-9 and J1-10 shall be open when interlock plate 10548925 is not mounted on klystron power supply. When interlock plate is properly mounted, continuity shall exist between J1-9 and J1-10.

3.5.2 Standby operation. With 28 ± 0.1 volts, direct current (Vdc) applied between J1-A3(+) 1 /, J1-4(+) and J1-A5(Ret), current drawn by klystron power supply shall be less than 2.8 amperes (A). Standby operation (maximum current drain) shall occur 20 ± 3 seconds (s) after voltage is applied.

3.5.2.1 Filament voltage. Voltage between J2-B(-) and J2-C(Ret) shall be 5.2 ± 8:$^{3}_{2}$~ Vdc under the following conditions:

a. As specified in 3.5.2.
b. 2 ohm, 50 watt (W) resistive load between J2-B and J2-C.
c. Shorting wire between J1-A5, J3, and chassis ground.
d. Filament fuse continuity between J2-F and J2-B.

Voltage output between **J2-E(-) and** J2-C(Ret)shall be 4.3 **Kilovolts, direct current (KVdc) in standby operation as specified in** *3.5.2.*

1/ **The information in parentheses refers to the polarity applied to this terminal. This notation shall be applied throughout this specification.**

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3.5.2.3 Uncompensated output ripple. Ripple at approximately 3.5 kilohertz (KHz) between J2-E(-) and J2–C(Ret) shall be less than 1.5 volts (V) peak–to-peak in standby operation as specified in 3.5.2.

3.5.3 Radiate operation. The klystron power supply shall be in radiate condition and the amount of current drawn by the klystron power supply shall be 8.0 ± 1.0 amperes dc under the following conditions:

a. 28 ± 0.1 Vdc applied between J1–4(+), J1–A3(+), and J1-A5(Ret).
b. 28 ± 0.1 Vdc applied between J1-A4(+) and J1-5(Ret).
c. Shorting wire between J1-5 and J1-A2.
d. Current input limited to 14 milliamperes (mA) applied to J1-11(+) and J1-A2(Ret).

3.5.3.1 Modulator dc voltage. Voltage between TP1(-) and TP-2 **(Ret) shall be 10.7 + 1.7 Vdc under conditions specified in 3.5.3. Voltage between TP2(+) and TP4(Ret) shall be 1.9 ± 0.3 Vdc under conditions specified in 3.5.3.**

3.5.3.2 **Square wave output. Signal between J1-13 and J1-14 shall be a square wave as described in the following under the conditions specified in 3.5.3 and with a 93-ohm resistive load** between **J1-13 and J1-14:**

a.	**Amplitude**	**195 ± 10 V peak-to-peak**
b.	**Frequency**	**305 ± 1 KHz**
c.	**Ripple on peaks -**	**less than 1 V peak-to-peak (Not including high frequency spikes)**

3.5.3.3 Output Pulse waveform. The waveform shown in Figure 1 shall be present between J2-E(signal) and J2-C(Ret) under the following conditions:

a. As specified in 3.5.3.
b. Total of 25 picofarads (pF) capacitive load comprising lead and connector capacitance as well as lumped capacitance between J2-E and J2-C.
c. 0.6 microsecond (us), 18 volt, 20 KHz pulse applied to J1-A1.

3.5.3.4 **High voltage output. The waveform shown in Figure 1 shall be present between J2-E(signal) and J2-C(Ret) and the most** negative portion of the waveform shall not be greater than -15.5 KVdc under the following conditions:

a. As specified in 3.5.3(a) and (b).
b. 25 pF capacitive load between J2-E and J2-C.
c. 0.6 us, positive 18 volt, 20 KHz pulse applied to J1-A1 .
d. 9.0 +0.5/-0.0 KVdc applied between J3(-) and J1-15(Ret).

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3.5.4 overload circuitry.

3.5.4.1 High voltage overload. When klystron **power supply is** operating in the radiate condition as specified **in 3.5.3.3, and** a current input, limited to 25 ± 1 milliamperes (mA) **is applied** between J1-11(+) and J1-A2(Ret), the high voltage **overload circuitry** shall operate and cause the following:

a. **28 Vdc available between J1-1(+) and J1-A2(Ret).**
b. **Square wave specified in 3.5.3.2 not available between J1-13 and J1-14.**
c. **Waveform of Figure 1 not available between" J2-E(Signal) and J2-C(Ret).**

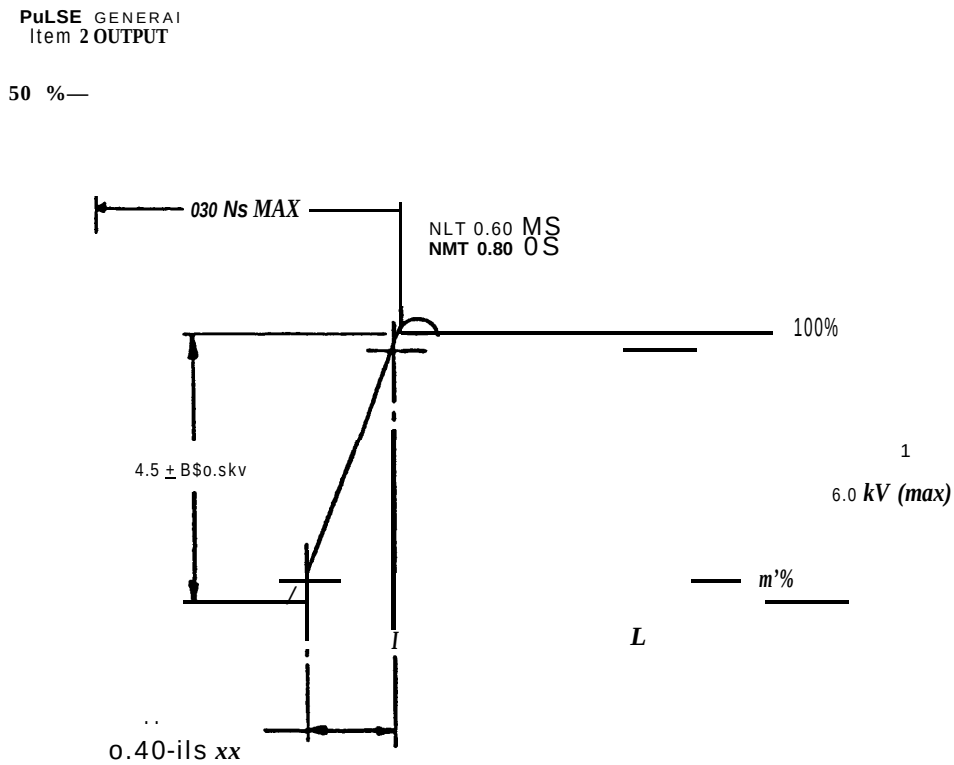


FIGURE 1. Modulator pulse voltage waveform.

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3.5.4.1.1 High voltage insufficient load. When the klystron power supply is operating in the radiate condition as specified in 3.5.3.3 and a current input of 5 ± 1 mA (insufficient load) is applied between J1-11(+) and J1-A2(Ret), the high voltage overload circuitry shall be as specified in 3.5.4.1 a, b, c.

3.5.4.2 Modulator overload. When klystron power supply is operating in the radiate mode as specified in 3.5.3.3. and a current input limited to 350 mA, is applied between TP-2(+) and J1-A2(Ret) the modulator overload circuitry shall operate and cause the following:

a. 28 Vdc available between J1-3(+) and J1-A2(Ret).
b. Square wave specified in 3.5.3.2 **not available between J1-13 and J1-14.**
c. Waveform of Figure 1 not available between J2-E(Signal) and J2-C(Ret).

3.5.4.3 overload reset. Reset circuitry shall operate in both overload circuits in either standby or radiate condition, when 28 Vdc is applied to J1-2(+) and **J1-A2(Ret)**.

3.5.4.4 Crowbar overload.

3.5.4.4.1 When the klystron power supply is operating in the standby mode as specified in 3.5.2, the crowbar circuitry shall operate causing the 28 Vdc supply to current limit when the 28 Vdc supply is within the range of +33.o Vdc to +35.o Vdc.

3.5.4.4.2 When the klystron power supply is operating in the radiate mode as specified in 3.5.3.3, the crowbar circuitry shall operate **causing the 28 Vdc supply to current limit when the current flow through z2 is within the range of 2 mA to 5 mA.**

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in **accordance with** the **technical provisions herein (see) 4.3).**

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (Task F) of paragraph **1.2 of MIL-STD-2000.**

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the **performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to**

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies **and services** conform to prescribed requirements.

4.1.1 Responsibility for compliance. **All** items must meet all requirements of sections 3 and 5 and all applicable drawings. **The inspections set forth in this specification** shall become a part of the-contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. **Special sampling inspection (see 4.5).**

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions: **e**

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three klystron power supplies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional tests) and Table II (Special sampling) herein and to any or all requirements of the applicable drawings.

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4.3.3 **Rejection. If any assembly, component or test specimen fails to** comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components. assemblies and subassemblies shall be inspected in accordance-with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 **Final acceptance inspection. Subsequent to first** article approval, examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot m

4.4.2.3 Functional tests. The tests in Table 1 shall be performed on a 100% basis.

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TABLE I. Functional tests
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	**4.7.2**
102. Performance	3.5	**4.7.3**
Minor:	None	**N/A**

4.5 Special sampling. One klystron power supply out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall **have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.**

TABLE II. Special samples

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
Critical:	**None**	N/A
Major:		
103. **Vibration**	3.3.1	4.7.1.1
104. shock	3.3.2	4.7.1.2
105. Altitude	3*3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3*5	4.7.1.5
Minor:	None	**N/A**

4.5.1 **Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.**

4.6 Inspection equipment.

4.6.1 **Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.**

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3. .

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE 111. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Power supply	Dc , 28 ± 0.1 V at 10 A.
2.	Pulse generator	Capable of providing 0.6 us, positive 18 volt pulse into 50 ohms at 20 KHz.
3.	Resistive load	93 ohms ± 10%, at 150 W.
4.	Power supply	Current limited, 10 to 350 MA.
5.	Resistive load	2 ohms ± lo%, 50 w.
6.	Capacitive load	25 pF, 6 KV.
7.	Power supply	9.5 KV at 10 mA, ripple on output less than 0.01% rms, short circuit protected.
8.	Voltmeter	0 to 30 Vdc, accurate of 1%, 20,000 ohms per volt.
9.	Voltmeter	Electrostatic, 4 to 15.5 KVdc, accuracy of 1%.
10.	Oscilloscope	Input sensitivity: 0.05 to 50 V per division. Horizontal deflection: 10 us per division. Input impedance Including probe: 100 megohms. Probe attenuation: Xl000. Vertical passband: dc to 10 MHZ.
11.	Ammeter	Dc , 0-10 A, accuracy of 1%.
12.	Switch	(a) Single pole, single throw, 3A capacity. (b) Single pole, single throw, 10A capacity. (c) Momentary contact, normally open, 3 A capacity.
13.	Indicator lamp and socket	28 volt, incandescent. (3 each)
14.	Shorting wire	Wire, insulated.
15.	Inductor	3.7 MH at 10 A, 50 WVDC (10548569)

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TABLE III. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION	
16.	Diode	JAN 1N5619A
17.	Test resistor	1 Megohm

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4*7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The klystron power supply shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the klystron power supply shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock . This test shall be performed as specified in Procedure 1, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, Klystron Power Supply shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (Operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

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4.7.1.4.1 High (Operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL=STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliante with 3.3.4.

4.7.1.5 Humidity (operating). This test shall be performed in accordance with procedure III, Method 507 of MIL-STD-810. The unit shall be operated in radiate mode for a minimum of 15 minutes during the performance testing of Step 7.

4.7.2 Electrical. Centinuity shall beinspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected klystron power supplies shall be tested as follows: All following tests except those specified in 4.7.3.3.4 shall be performed with test equipment connected as shown in Figure 2 with referenced items of test equipment from Table III.

WARNING: Voltages dangerous to life are present during tests specified in this section. Death on contact may result from failure to observe safety precautions and good high voltage practices.

4.7.3.1 Interlock switch. Verify that Switches Sl, S2 and S3 shown in Figure 2 are open. Apply 28 Vdc ± 0.1 Vdc between J1-10(+) and Jl-15(Ret) using power supply (Table III, item 1). Operate interlock switch by mounting interlock plate, 10548925, on klystron power supply. Verify that interlock lamp shown in Figure 2 is illuminated. Remove interlock plate. Verify that interlock lamp is extinguished.

4.7.3.2 Standby operation. Operate switch S1, item 12(a) to standby position. Using power supply (Table III, item 1) apply 28 ± 0.1 Vdc between J1-A3(+), J1-4(+), and J1-A5(Ret). Using ammeter (Table III, item 11), verify compliance with 3.5.2. Allow unit to remain in standby.

4.7.3.2.1 Filament voltage. Using voltmeter (Table III, item 8), verify compliance with 3.5.2.1.

4.7.3.2.2 Bias voltage. Using voltmeter (Table III, item 9), verify that voltage between J2-E(-) and J2-C(Ret) is in compliance with 3.5.2.2. Record voltage for further use.

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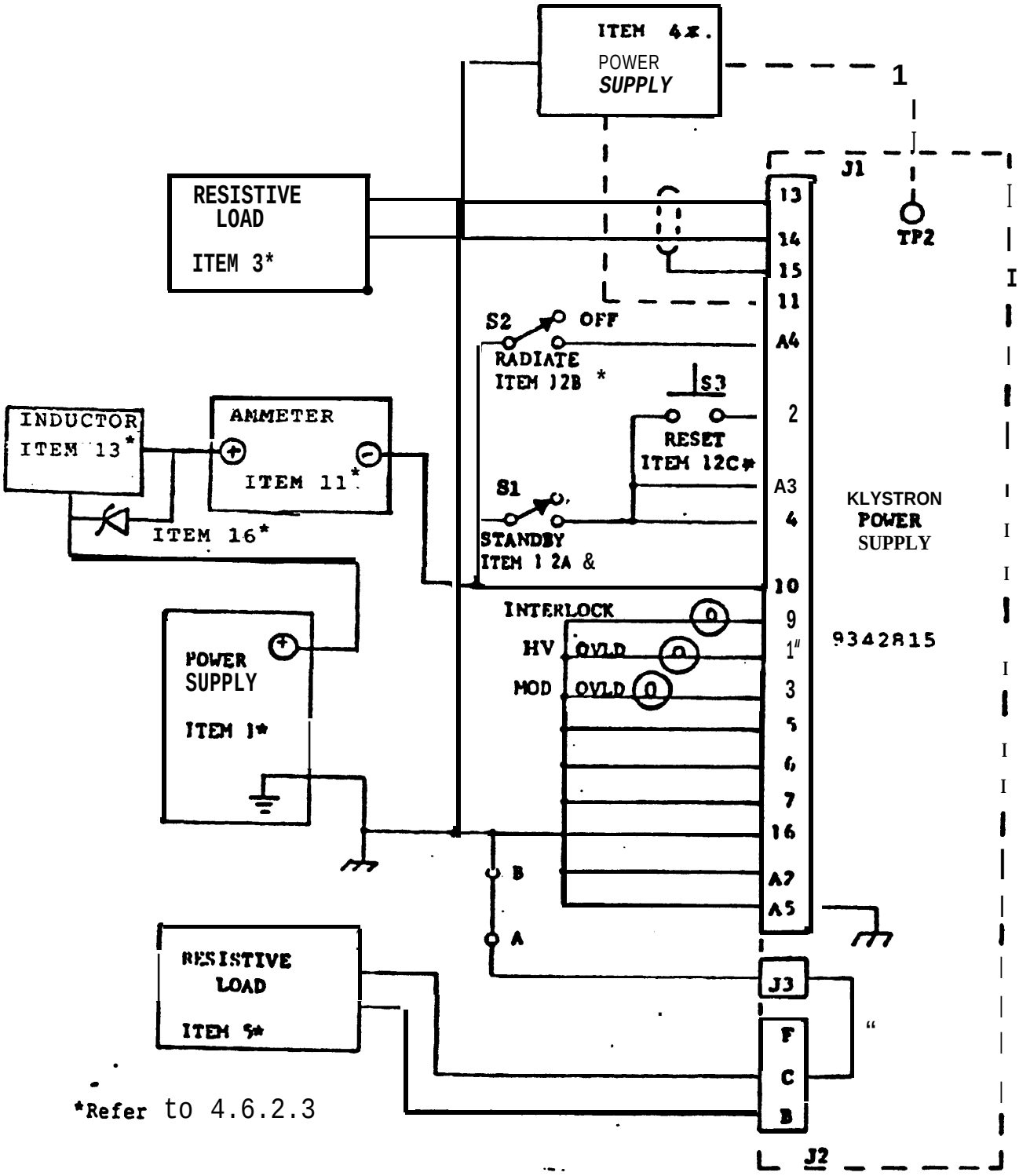


FIGURE 2. Klystron power supply test setup.

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4.7.3.2.3 Uncompensated output ripple. Using oscilloscope (Table III, item 10), verify compliance with 3.5.2.3.

4.7.3.3 **Radiate operation. Operate switch S1 to standby. After 23 seconds, operate switch S2 to radiate. Using ammeter (Table III, item 11), verify that current drawn by klystron power supply is 8.0 ± 1.0 amperes. Allow unit to remain in radiate operation.**

4.7.3.3.1 Modulator dc voltage. Use voltmeter (Table III, item 8) to verify compliance with 3.5.3.1.

4.7.3.3.2 **Square** wave output. Use oscilloscope (Table **III, item** 10) without high voltage probe to verify compliance with ***3.5.3.2.***

4.7.3.3.3 Output pulse waveform. Use pulse generator (Table III, item 2), apply a 0.6 microsecond, positive 18 volt, 20 kilohertz pulse to J1-A1. Verify, using oscilloscope (Table 111, item 10) with high voltage probe, that pulse at J2-E(signal) and J2-C(Ret) appears as shown in Figure 1. Using voltmeter (Table III, item 9), verify that voltage between J2-E(-) and J2-C(Ret) is greater than **voltage value recorded in 4.7.3.2.2.**

4.7.3.3.4 High voltage output. **Remove all voltages from unit under test. Remove shorting wire between points A and B in Figure 2. Using power supply (Table 111, item 7), apply 9+0.5/-o KVdc between point A(-) and point** B(Ret). Turn on standby switch S1. Using pulse generator (Table III, item 2), apply a 0.5 micro-second, positive 18 volt, 20 kilohertz pulse to J1-A1. After 23 seconds, turn on switch S2. Using oscilloscope (Table III, item 10), verify that signal between J2-E(signal) and J1-15(ground) conforms to waveform in Figure 1 and most negative portion of pulse is no greater than -15.5 KVdc. Operate switches S1 and S2 to off position. Turn off high voltage power supply (Table III, item 7) and disconnect from test setup. Replace shorting wire between points A and B.

4.7.3.4 overload circuitry.

4.7.3.4.1 **High voltage overload. Operate switch S1 to standby. After 23 seconds, operate switch S2 to radiate. Using pulse** generator (Table III, item 2), apply a 0.6 microsecond positive 18 volt, 20 kilohertz pulse to J1-A1. Using oscilloscope (Table III, item 10), verify that waveform shown in Figure 1 is present between J2-E(signal) and J2-C(ground). Using power supply (Table III, item 4), apply an input, current limited to 25 ± 1 mA in the high voltage overload condition and 5 ± 1 mA in the high voltage insufficient load condition, between J1-11(+) and J1-A2(Ret). Verify that high voltage pulse is illuminated. Turn off switch S2, remove pulse from J1-11 and momentarily depress reset switch S3. Verify that high voltage overload lamp is extinguished. operate switch S2 to radiate position. Verify that high voltage pulse-appears on oscilloscope and high voltage overload lamp is not illuminated.

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4.7.3.4.2 Modulator power supply overload. Operate switch S1 to standby. After 23 seconds, operate switch S2 to radiate. Using pulse generator (Table III, item 2), apply a 0.6 microsecond, positive 18 volt, 20 kilohertz pulse to J1-A1. Verify, using oscilloscope (Table III, item 10), that waveform shown in Figure 1 is present between J2-E(Signal) and J2-C(Ret). Using power supply (Table III, item 4), apply an input, current limited to 350 mA, between TP2(+) and J1-A2(Ret). Verify that high voltage pulse is no longer present at J2-E and modulator overload light is illuminated. Turn off switch S2, **remove pulse from TP2 and momentarily depress reset switch S3. Verify that modulator overload lamp is extinguished. Operate switch S2 to radiate position. Verify that high voltage pulse appears on oscilloscope and modulator overload lamp is not illuminated.**

4.7.3.4.3 28 Vdc overvoltage test. Operate switch S1 to standby. Wait 23 seconds then monitor the voltage between J1–A3 and **J1-16 of the** klystron power supply. Slowly increase the 28 Vdc power supply voltage (Table 111, Item 1) until the power supply is current limited. The voltage at J1-A3 shall be 34.0 Vdc ± 1 Vdc. Turn S1 to off. Reset power supply (Table II, Item 1) to 28 Vdc ± .1 vdc.

4.7.3.4.4 Grid bias overload test. Connect one end of the 1 Megohm test resistor (Table III. Item 14) to test fixture chassis. **Turn-switch S1 to standby. Wait 20 seconds. Turn switch S2 to radiate. Carefully touch the other end of the test resistor to the circuitry that connects to J2-E. The** 28 **Vdc power supply (Table III, Item 1) shall go into current limiting. Remove the test resistor. Turn** S1 **and** S2 **to off. Turn S1 to on and wait for the 20 seconds standby mode. Turn on *S2* and the klystron power supply shall go into radiate mode. Turn** switch S1 and S2 to off.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS10549202. The level of protection shall be as specified in the procurement document.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory-)

6.1 Intended use. The klystron power supply is used in Test Set, Subassembly MX-8047/TPM-22, part of Radar Test Set AN/TPM-22A. The klystron power supply is identical to, and simulates the performance of, a subassembly of Radar Set AN/VPS-2. output

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voltage from the klystron power supply is used to check performance of the transmitter of Radar Set AN/VPS-2. The klystron power supply is also used in Receiver Transmitter RT-680/VPS-2, part of Radar Set AN/VPS-2, to provide high voltage for the transmitter.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of Conformance for each lot or shipment of products.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on-the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 High voltage warning. High voltage dangerous to life are present when the klystron power supply is under test. Death on contact may result from failure to observe safety precautions and good high voltage practices.

6.6 Subject term (key word) listing

Test Set, Subassembly MX-8047/TPM-22
Radar Set AN/TPM-22A
Receiver Transmitter RT-860/VPS-2
Radar Set AN/VPS-2

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6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing activity:
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(Project 1285-A701)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
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3. DOCUMENT TITLE

POWER SUPPLY: 9342815 (KLYSTRON)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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